Software for Attendance Check for Staff Using Geofencing Technology

SENIOR PROJECT BL1-2026







What is Our Project?

A mobile-based attendance management system that uses geofencing technology to detect when university staff enter or leave a predefined faculty area.

Key Features:

- GPS-based location detection
- Automatic check-in/check-out notifications
- Manual attendance confirmation
- Automatic attendance after a grace period
- Centralized attendance records
- Administrator dashboard



Problem Statement & Proposed Solution



Problems

- Staff forget to check in or out
- Manual attendance is inconvenient
- Missing records require corrections
- GPS readings may be inaccurate
- Immediate automatic attendance can create errors



Our Solution

- Detect staff arrival and departure
- Send check-in/out notifications
- Allow manual confirmation
- Auto-record after a grace period
- Revalidate GPS location before automatic attendance

How Geofencing Works

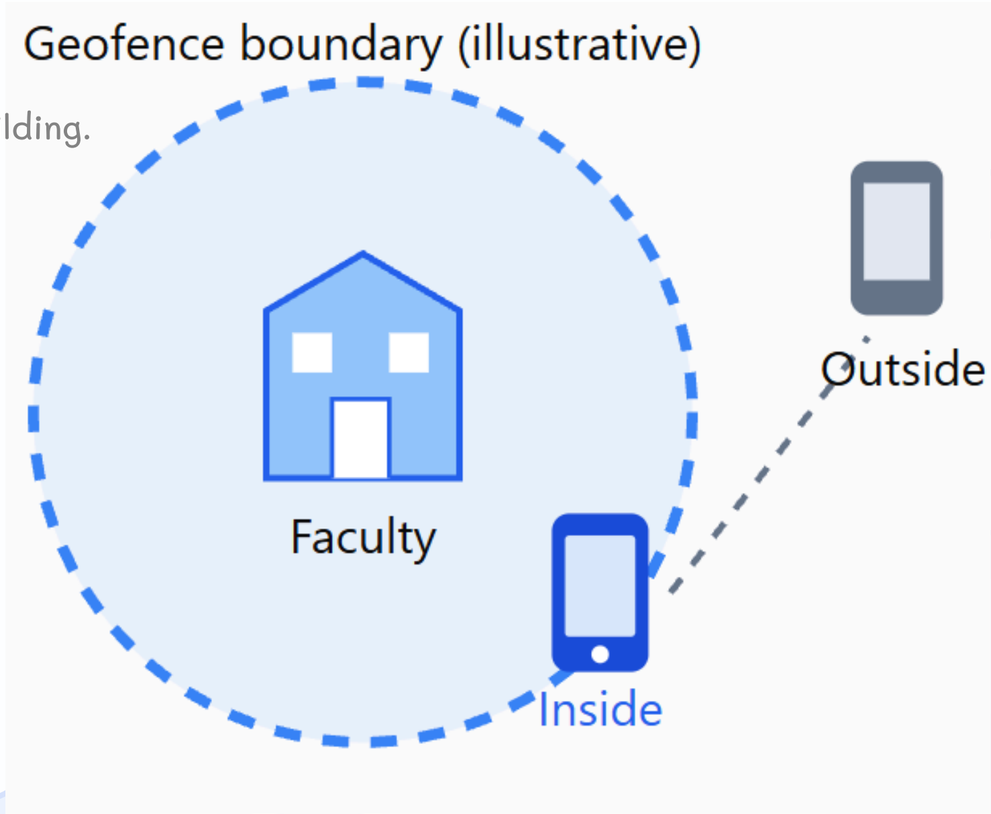
Geofencing Technology

A virtual geographic boundary is established around the faculty building.

- Smartphone provides location coordinates
- System detects entry and exit events
- Proposed initial radius: 150 meters
- Radius is configurable and will be tested
- Location is revalidated before automatic attendance

Inside: Distance ≤ Geofence Radius

Outside: Distance > Geofence Radius



Requirements Specification

Functional Requirements

ID	Requirement
FR01	User login and authentication
FR02	Detect entry/exit of geofence
FR03	Send check-in/out notifications
FR04	Support manual attendance confirmation
FR05	Automatically check in/out after timeout and location validation
FR06	Store attendance records in database
FR07	Display attendance history
FR08	Provide administrator dashboard
FR09	Allow authorized attendance corrections

User Roles & Permissons

Three User Roles

1. **Staff Member**

Login, receive notifications, check in/out, view attendance history.

2. **Attendance Administrator**

View staff attendance, correct records, generate reports.

3. **System Administrator**

Manage accounts, user roles, geofence configurations, and system settings.

Backend, Database & Non-Functional Requirements

Backend Responsibilities

- User authentication and role-based authorization
- Attendance processing and validation
- Duplicate record prevention
- Configuration and notification tracking
- Audit logging

Database Tables

- Users
- Attendance Records
- Geofence Configuration
- Notification Events
- Audit Logs

Non-Functional Requirements

- Secure authentication and communication
- Reliable background location detection
- Appropriate data privacy and access control
- Consistent attendance records
- Support for location uncertainty and errors





Proposed System Architecture

How the Components Connect

Staff Mobile Application

Login • Location detection • Notifications • Attendance history

↓↑ Secure API communication

Backend Server

Authentication • Attendance decisions • Validation • Access control

↓↑ Read / Write

Attendance Database

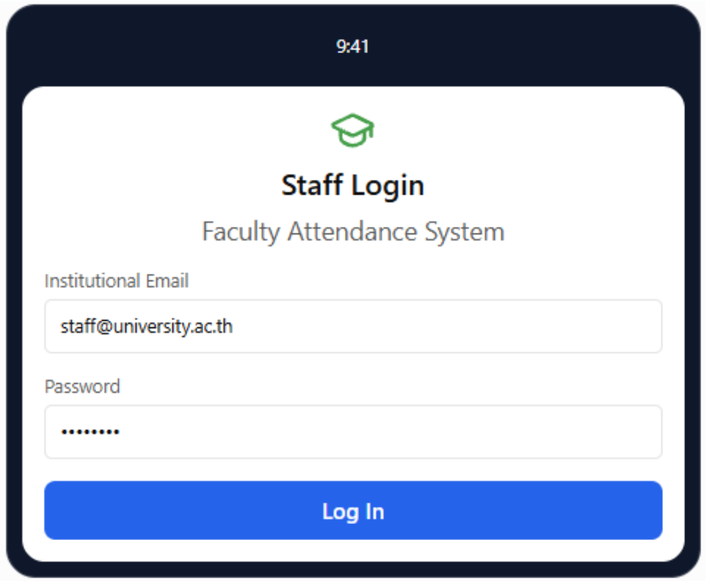
Users • Attendance • Notifications • Audit logs

Administrator Web Dashboard

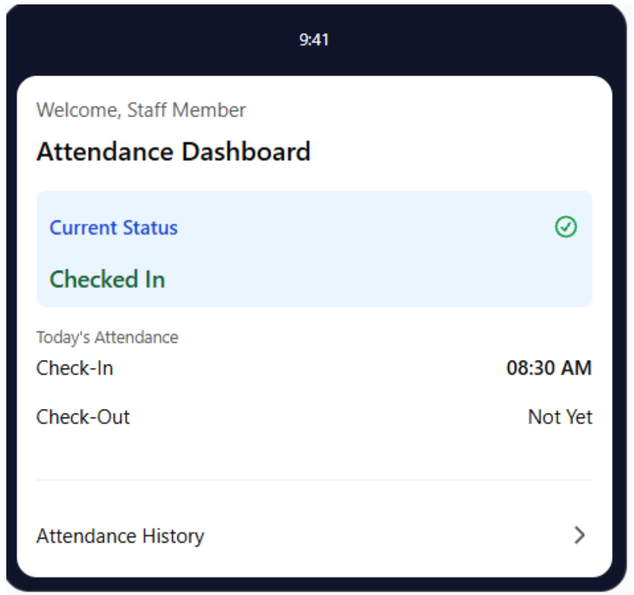
Connects securely to the backend API to review and manage attendance

UI Mockups: Staff Mobile Application

Proposed User Interface



Mockup A: Login Screen



Mockup B: Staff Dashboard

Attendance Workflow Demonstration

1. Staff enters the geofence

Phone detects an entry event

↓

2. Check-In notification appears

Staff can manually confirm attendance

↓

3. Did the staff confirm?

YES	NO
Manual Check-In	Wait for grace period

↓

4. Revalidate Location

If unanswered, check whether staff remains inside

↓

5. Record Attendance

Automatic Check-In only if location is valid and inside; otherwise do not auto-record

1. Staff exits the geofence

Phone detects an exit event

↓

2. Check-Out notification appears

Staff can manually confirm departure

↓

3. Did the staff confirm?

YES	NO
Manual Check-Out	Wait for grace period

↓

4. Revalidate Location

If unanswered, check whether staff remains outside

↓

5. Record Attendance

Automatic Check-Out only if location is valid and outside; otherwise do not auto-record

GROUP MEMBERS

Thanyawach Keawhan
6622780094

Nontapat Chaopratoom
6622772489

Suleyman Hannyyev
6622790291